8.3 Analyzing Geometric Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.3 (30-minute plan) · Student sheet

I can…

- tell whether a sequence is geometric and write its rule.
- write a rule from one term and the common ratio, or from two terms.
- find the sum of a finite geometric series.

Vocabulary · fill in the missing word

Geometric sequence — the ratio of _____ terms is constant

Common ratio r — the _____ ratio

Rule — a_n = _____

Sum of a finite geometric series — S_n = _____, $r \neq 1$

Part 1 · Notes and examples (fill in each note, then work the example)

Geometric sequence: Geometric: the _____ of consecutive terms is constant (r).

Example 1

Tell whether each sequence is geometric.

a. 5, 10, 20, 40, … b. 1, 4, 9, 16, … c. 96, −48, 24, −12, …

Rule: Rule: a_n = _____

Example 2

Write a rule for the nth term of 4, 20, 100, 500, … Then find a_7.

Sum of a finite geometric series: Sum: S_n = _____

Example 3

Find Σ (i = 1 to 8) $4(3)^{i-1}$.

Part 2 · Partner practice

Partner practice

One term of a geometric sequence is $a_4 = 12$, and $r = 2$. Write a rule for the nth term.